

Assignment -4

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Decrypt the following ciphertext using the Hill cipher.

1. Cipher text : HLDTOIHJD XWQCMAG

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Use ($A=0, B=1, \dots, Z=25$).

Find the plain text.

Step 1: Find determinant of K

$$\det(K) = 3(7) - 5(2) = 21 - 10 = 11.$$

Step 2: Find K^{-1}

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$11^{-1} \pmod{26} : \quad 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$11^{-1} \pmod{26} = 19 \pmod{26} = \underline{\underline{19}}$$

$$\therefore K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

Step 3: Divide ciphertext into pairs

HL DT QI HJ DX WQ CM AG

Formula: $P = K^{-1}C \pmod{26}$

$$HL = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{matrix} T \\ O \end{matrix}$$

$$DT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$QI = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{matrix} E \\ C \end{matrix}$$

$$HJ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$DX = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$WQ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} E \\ S \end{matrix}$$

$$CM = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{matrix} S \\ A \end{matrix}$$

$$AG = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{matrix} G \\ E \end{matrix}$$

$\therefore$ Plaintext = TOP SECRET MESSAGE.

2. Consider the Hill Cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

cipher text : RXAODY

Find the plaintext for the above cipher text.

Step 1: Find determinant of K

$$\det(K) = 24(14) - 19(5) = 336 - 95 = 241$$

$$241 \bmod 26 = 7$$

Step 2: Find K^{-1}

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$(\det K)^{-1} = 7^{-1} : 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[26 + 7(-3)] + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} \bmod 26 = 15 \bmod 26 = 15$$

$$\therefore K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

Step 3: Divide cipher text into pairs;

RX AO DY

$$\text{Formula: } P = K^{-1} C \pmod{26}$$

$$\cancel{RX} = \cancel{\begin{bmatrix} 17 \\ 23 \end{bmatrix}} \quad RX = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 57 \\ 557 \end{bmatrix} \pmod{26} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \pmod{26} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 4 \\ 17 \end{bmatrix}$$

$$= \begin{matrix} E \\ R \end{matrix}$$

FL OW ER

$\therefore$ Plain text = FLOWER.

Confusion vs. Diffusion

Analyze the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext.

Cipher	Does it change the character / symbol value?	Does it change the character position?	Confusion / Diffusion	Justification
Shift Cipher	Yes	No	Confusion	Each plaintext character is replaced by another character by shifting its value. The position remains unchanged.
Multiplicative Cipher	Yes	No	Confusion	Each character value is multiplied by a key modulo 26. Character positions remain unchanged.
Affine Cipher	Yes	No	Confusion	Each character is transformed using $C = (aP + b) \bmod 26$. The symbol value changes but its position does not.
Transposition Cipher	No	Yes	Diffusion	The character themselves remain unchanged, but their positions are rearranged.

Based on your observations, explain the relationship between substitution and confusion and between transposition and diffusion.

- Relationship between substitution and confusion:

- * Substitution ciphers such as shift, Multiplicative and Affine ciphers replace one symbol with another.

- * Therefore, they primarily provide confusion by making the relationship between the plaintext symbols and ciphertext symbols less direct.

- Relationship between transposition and Diffusion:

- * A transposition cipher does not change the actual characters. Instead, it rearranges their positions.

- * This spreads the plaintext characters across different positions in the ciphertext and therefore provides diffusion.

- Substitution $\rightarrow$ Confusion

- Transposition $\rightarrow$ Diffusion

Transposition

Encryption

Encrypt the message "the arrow is pointing north direction" using

- simple with key = 4

→ Key = 4

∴ No. of columns = 4

T	H	E	A
R	R	O	W
I	S	P	O
I	N	T	I
N	G	N	O
R	T	H	D
I	R	E	C
T	I	O	N

∴ Ciphertext = TRIINRITHRSNGTRI EOPTNHEDAWOI ODCN.

- keyword with keyword = ATLAS

Keyword length = 5 ∴ No. of columns is 5

A	T	L	A	S
1	5	3	2	4
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

∴ Ciphertext = TRPIRRO AINNDTX EWIGHCX RSTOIIXHOONTEN

- double

Key 1 : BREAK

Key 2 : log

→ Key 1 = BREAK, length = 5, No. of columns = 5

B 2	R 5	E 3	A 1	K 4
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

Initial ciphertext = AINNDTXTRPIRROEWIGHCX RSTOIIXHOONTEN

Key 2 = LOG, length = 3, No. of columns = 3

L 2	O 3	G 1
A	I	N
N	D	T
X	T	R
P	I	R
R	O	E
W	I	G
H	C	X
R	S	T
O	I	I
X	H	O
O	N	T
E	N	X

∴ Final ciphertext is

NTRREGXTIOTX ANXPRW HROXOE IDTIOICSIHNN

Decryption

- simple with key 10

TIDIE MRNME REOGO SANOW RSLTP EERDO SSSNU AHTUD
BIENP GODAE PEIXD ELNSX

→ String length = 60, Key = 10

$$\therefore \frac{60}{10} = 6 \Rightarrow \text{No. of rows} = 6$$

T	R	O	O	P	S	H	E	A	D
I	N	G	W	E	S	I	N	E	E
D	M	O	R	E	S	U	P	P	L
I	E	S	S	E	N	D	G	E	N
E	R	A	L	D	U	B	O	I	S
M	E	N	T	O	A	I	D	X	X

∴ Plaintext = TROOPS HEADING WEST NEED MORE SUPPLIES
SEND GENERAL DUBOIS MEN TO AID.

- keyword

SECRET

Ciphertext: ILEDT NHPTR SDIET MAXSA XERXT ANONI
SNFOT X

→ Key = SECRET, Key length = 6, No. of columns = 6

$$\text{Ciphertext length} = 36 \therefore \text{No. of rows} = \frac{36}{6} = 6$$

S E C R E T

5 2 1 H 3 6 → Numbering of the keyword.

S	E	C	R	E	T
5	2	1	H	3	6
T	H	I	S	I	S
A	P	L	A	I	N
N	T	E	X	T	F
O	R	D	E	M	O
N	S	T	R	A	T
I	O	N	X	X	X

∴ Plaintext : THIS IS A PLAINN TEXT FOR DEMONSTRATION.

- keyword

keyword - PLAIN

CIPHER - VEY EAX ARA ATX HGD

→ Keyword = PLAIN, length = 5, No. of columns = 5

Ciphertext length = 15, No. of rows = $\frac{15}{5} = 3$

P	L	A	I	N
5	3	1	2	4
H	A	V	E	A
G	R	E	A	T
D	A	Y	X	X

∴ Plaintext is

HAVE A GREAT DAY

Permutation :

Find the inverse permutation for

1. 4 3 1 5 2

Position	1	2	3	4	5
P-box	4	3	1	5	2

Inverse Permutation	3	5	2	1	4
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$\therefore$ Inverse Permutation, $P^{-1} = (3, 5, 2, 1, 4)$

2. 7 4 1 3 5 2 6

Position	1	2	3	4	5	6	7
P-box	7	4	1	3	5	2	6
Inverse Permutation	3	6	4	2	5	7	1

$\therefore$ Inverse Permutation, $P^{-1} = (3, 6, 4, 2, 5, 7, 1)$

3. 4 1 3 2

Position	1	2	3	4
P-box	4	1	3	2
Inverse Permutation	2	4	3	1

$\therefore$ Inverse Permutation, $P^{-1} = (2, 4, 3, 1)$

4. 3 5 1 4 2

Position	1	2	3	4	5
P-box	3	5	1	4	2
Inverse Permutation	3	5	1	4	2

∴ Inverse Permutation, $P^{-1} = (3, 5, 1, 4, 2)$